

## Lab - Calculate IPv4 Subnets

### Objectives

Part 1: Determine IPv4 Address Subnetting

Part 2: Calculate IPv4 Address Subnetting

### Background / Scenario

The ability to work with IPv4 subnets and determine network and host information based on a given IP address and subnet mask is critical to understanding how IPv4 networks operate. The first part is designed to reinforce how to compute network IP address information from a given IP address and subnet mask. When given an IP address and subnet mask, you will be able to determine other information about the subnet.

- 1 PC (Windows with Internet access)
- Optional: IPv4 address calculator

### Instructions

Fill out the tables below with appropriate answers given the IPv4 address, original subnet mask, and new subnet mask.

#### Problem 1:

| Given:               |                 |
|----------------------|-----------------|
| Host IP Address:     | 192.168.200.139 |
| Original Subnet Mask | 255.255.255.0   |
| New Subnet Mask:     | 255.255.255.224 |

| Find:                                     |                 |
|-------------------------------------------|-----------------|
| Number of Subnet Bits                     | 3               |
| Number of Subnets Created                 | 8               |
| Number of Host Bits per Subnet            | 5               |
| Number of Hosts per Subnet                | $32 - 2 = 30$   |
| Network Address of this Subnet            | 192.168.200.128 |
| IPv4 Address of First Host on this Subnet | 192.168.200.129 |
| IPv4 Address of Last Host on this Subnet  | 192.168.200.158 |
| IPv4 Broadcast Address on this Subnet     | 192.168.200.159 |

**Problem 2:**

| Given:               |               |
|----------------------|---------------|
| Host IP Address:     | 10.101.99.228 |
| Original Subnet Mask | 255.0.0.0     |
| New Subnet Mask:     | 255.255.128.0 |

| Find:                                     |                              |
|-------------------------------------------|------------------------------|
| Number of Subnet Bits                     | 9                            |
| Number of Subnets Created                 | $2^9 = 512$                  |
| Number of Host Bits per Subnet            | 15                           |
| Number of Hosts per Subnet                | $2^{15} = 32768 - 2 = 32766$ |
| Network Address of this Subnet            | 10.101.0.0                   |
| IPv4 Address of First Host on this Subnet | 10.101.0.1                   |
| IPv4 Address of Last Host on this Subnet  | 10.101.127.254               |
| IPv4 Broadcast Address on this Subnet     | 10.101.127.255               |

**Problem 3:**

| Given:               |               |
|----------------------|---------------|
| Host IP Address:     | 172.22.32.12  |
| Original Subnet Mask | 255.255.0.0   |
| New Subnet Mask:     | 255.255.224.0 |

| Find:                                     |                            |
|-------------------------------------------|----------------------------|
| Number of Subnet Bits                     | 3                          |
| Number of Subnets Created                 | 8                          |
| Number of Host Bits per Subnet            | 13                         |
| Number of Hosts per Subnet                | $2^{13} = 8192 - 2 = 8190$ |
| Network Address of this Subnet            | 172.22.32.0                |
| IPv4 Address of First Host on this Subnet | 172.22.32.1                |
| IPv4 Address of Last Host on this Subnet  | 172.22.63.254              |
| IPv4 Broadcast Address on this Subnet     | 172.22.63.255              |

**Problem 4:**

| Given:               |                 |
|----------------------|-----------------|
| Host IP Address:     | 192.168.1.245   |
| Original Subnet Mask | 255.255.255.0   |
| New Subnet Mask:     | 255.255.255.252 |

| Find:                                     |                   |
|-------------------------------------------|-------------------|
| Number of Subnet Bits                     | 6                 |
| Number of Subnets Created                 | $2^6 = 64$        |
| Number of Host Bits per Subnet            | 2                 |
| Number of Hosts per Subnet                | $2^2 = 4 - 2 = 2$ |
| Network Address of this Subnet            | 192.168.1.244     |
| IPv4 Address of First Host on this Subnet | 192.168.1.245     |
| IPv4 Address of Last Host on this Subnet  | 192.168.1.246     |
| IPv4 Broadcast Address on this Subnet     | 192.168.1.247     |

**Problem 5:**

| Given:               |               |
|----------------------|---------------|
| Host IP Address:     | 128.107.0.55  |
| Original Subnet Mask | 255.255.0.0   |
| New Subnet Mask:     | 255.255.255.0 |

| Find:                                     |                       |
|-------------------------------------------|-----------------------|
| Number of Subnet Bits                     | 8                     |
| Number of Subnets Created                 | 256                   |
| Number of Host Bits per Subnet            | 8                     |
| Number of Hosts per Subnet                | $2^8 = 256 - 2 = 254$ |
| Network Address of this Subnet            | 128.107.0.0           |
| IPv4 Address of First Host on this Subnet | 128.107.0.1           |
| IPv4 Address of Last Host on this Subnet  | 128.107.0.254         |
| IPv4 Broadcast Address on this Subnet     | 128.107.0.255         |

**Problem 6:**

| Given:               |                 |
|----------------------|-----------------|
| Host IP Address:     | 192.135.250.180 |
| Original Subnet Mask | 255.255.255.0   |
| New Subnet Mask:     | 255.255.255.248 |

| Find:                                     |                   |
|-------------------------------------------|-------------------|
| Number of Subnet Bits                     | 5                 |
| Number of Subnets Created                 | 32                |
| Number of Host Bits per Subnet            | 3                 |
| Number of Hosts per Subnet                | $2^3 = 8 - 2 = 6$ |
| Network Address of this Subnet            | 192.135.250.176   |
| IPv4 Address of First Host on this Subnet | 192.135.250.177   |
| IPv4 Address of Last Host on this Subnet  | 192.135.250.182   |
| IPv4 Broadcast Address on this Subnet     | 192.135.250.183   |

**Reflection Question**

Why is the subnet mask so important when analyzing an IPv4 address?